COSMICFORGE
REAL ESTATE

PAYMENT, WALLET & PROTECTED TRANSACTIONS

MASTER ARCHITECTURE & BUSINESS-RULE BLUEPRINT

Version 1.0 | 10 October 2026

Document control	Details
Owner	CosmicForge Tech Limited / CosmicForge Real Estate product leadership
Authority	Eight Real Estate Product Feature Documents v5.0, dated 22 July 2026, plus recorded stakeholder decisions
State	Baseline technical design approved for specification; financial-provider and legal release gates OPEN
Primary market	Nigeria first; other jurisdictions only through separately approved policy packs
Purpose	Comprehensive reference for Payment Service requirements, integration contracts, implementation phases and certification

This document specifies product intent and system design. It is not evidence of a licensed escrow service, a Paystack-approved use case, a live stored-value product or production-readiness.

Document authority and reading guide

This is the proposed controlled master blueprint for the shared CosmicForge Real Estate Financial Core. It consolidates the user-approved architectural decisions and the payment-related feature requirements of all eight product documents. It does not supersede the source documents: where a conflict is found, log it and obtain product/legal approval before changing either implementation or source requirements.

Status label	Interpretation
LOCKED — PRODUCT	A business/technical direction explicitly agreed in this conversation.
REQUIRED — SOURCE	Capability expressly specified by one or more source product documents.
DESIGN — ENGINEERING	Recommended design detail requiring architecture review and test evidence.
OPEN — BUSINESS	Rates, thresholds, default rules or authorities not yet approved.
GATED — EXTERNAL	Depends on written provider approval, licensed operating model or legal review.
DEFERRED	Intentionally outside this core implementation phase, with safe future interfaces reserved.

Source documents, all v5.0 — 22 July 2026

ID	Source	Payment-relevant chapters
00	CosmicForge Real Estate Platform Corporate Product Overview	§10 full cost; §11 protected payments; §12 deposits; §23 disputes; §27 journeys
01	Unified Property Marketplace	§11 tenancy; §16 purchase; §17 off-plan; §18–20 hospitality/payment; §29 financing
02	Individual Property Owner	§3 payout; §8 costs; §14 rent/deposit; §15 renewals; §19 sale; §20 short-let
03	Business Property Operations	§4 enterprise finance; §9–16 rent/sale/development/hospitality; §23 reporting
04	Property Manager	§5 permissions; §17 fees and payouts; §21 disputes
05	Artisan and Property Service Provider	§6 quotation; §8 evidence; §13 invoices, payments and earnings
06	Professional Property Services	§4 fees; §12 staged work; §16 fees/invoices/payouts
07	Platform Administration and Operations	§10 costs; §11 payments/payouts/reconciliation; §12 deposits; §13 fraud; §15 disputes

The existing Individual Owner backend implementation specification and KarryGo Payment Service source audit are supporting engineering references, not overriding product authorities.

Executive decisions — locked baseline

- One authoritative Real Estate Payment Service and ledger serves every role and application; no independent per-app ledger.
- Every registered public user receives an internal wallet identity automatically; external virtual accounts require consent, eligibility and provider approval.
- Funds are collected, safeguarded and moved by approved regulated third parties. CosmicForge owns orchestrated payment logic, records, allocations, authorization and policies.
- Paystack is the preferred first integration, but the platform is provider-neutral at its domain boundary. Collection, custody, wallet funding, refunds and payout rails may differ.
- Paystack Manual Payouts is the preferred settlement arrangement where approved. Merchant balance alone is not proof of permissible buyer-seller escrow or user stored value.
- Beneficiary splits are computed and version-locked inside the CosmicForge ledger; unreleased shares are visible as pending/protected and never withdrawable.
- Withdrawal eligibility follows verifiable contractual milestones, dispute/lien holds, beneficiary validation, settled backing and wallet limits.
- All receiving user roles can withdraw eligible available earnings through an approved payout provider; funding and customer withdrawal are separately permissioned.
- Initial rental charges are separated from rent. Subsequent rent payments do not automatically repeat one-off platform/onboarding/legal fees.
- Recurring platform revenue defaults to a disclosed owner-side commission on rent collections (rate NOT fixed), or a separately approved owner subscription.
- Continuing/periodic tenancy and fixed-term agreement journeys are both supported; payment of the next rent installment does not automatically require a new contract.
- Customer wallet top-up and authorized spend are core; saved-bank-account auto-debit, card mandates and automated recurring rent scheduling are deferred to later phases.
- No go-live until approved legal/regulatory funds-handling structure, Paystack/provider commercial agreement, and financial certification are evidenced.

1. Scope, boundaries and operating principles

1.1 Financial Core ownership

- Wallet identity/accounting; durable payment intent and collection confirmation; double-entry books; settlement/custody reconciliation; calculation and reservation of beneficiary allocations.
- Contractual release policies; holds, disputed portions, release approvals; refunds, chargebacks, negative-balance controls; verified payout accounts and withdrawals.
- Provider registry/adapters; provider selection and liquidity; financial records, statements, receipts, audit trails, risk controls, finance-operations APIs and event publication.
- Recurring rental-obligation interface (not automatic saved-bank debit in v1); role-bound accounts and tax/fee breakdown metadata.

1.2 What other services own

Owning service	Authoritative facts Payment Core consumes
Marketplace	Payer identity/context, rental applications, booking selection, accepted all-in quote, customer consent, tenant/guest presence and dispute requests.
Individual Owner	Owner and property authority, tenancy/lease/offer, signed

Owning service	Authoritative facts Payment Core consumes
	version, move-in/out and inspection evidence, mandate restrictions, price approval.
Business Operations	Corporate legal owner, branch and staff approvals, developer project/phase, corporate asset and hospitality folios.
Property Manager	Active owner mandate, manager scope/rate/dates, property limits; cannot change owner beneficiary.
Artisan/Provider	Accepted quote, milestones, attendance, completion evidence, materials and approved extras.
Professional	Appointment/mandate, conflict check, signed report or deliverable, approved fee and completion.
Administration/Disputes	Approved finance policy, compliance decisions, fraud blocks, dispute case and release decision, dual-control authorizations.
Verification/Identity (shared)	Payer/beneficiary KYB/KYC badges and validity; no independent Payment Core assertion of identity truth.

1.3 Non-negotiable boundaries

- No app or human actor may directly update spendable money; only ledger-posted, policy-authorized operations change financial balances.
- A business event must be authenticated, source-bound, versioned, evidence-linked, and idempotent before affecting protected allocations.
- Payout beneficiary must be verified against legal owner or separately documented entitlement, not the listing creator or manager.
- No provider-scoped balances are assumed fungible across merchants, providers, legal entities, currencies or regulated custodial accounts.
- No claims that funds "never leave Paystack": in manual payout arrangements funds reside in a Paystack-identified commercial/banking structure, subject to settlement timing, terms and restrictions.
- Production transfer and fund-holding actions default to disabled pending commercial, regulatory and tested operational approval.

2. User wallet model — every role

LOCKED — PRODUCT: A wallet identity is created when an eligible user account is registered. A wallet is a ledger-based account view, not an individual bank account or guaranteed deposit-taking facility. The wallet must be able to display pending and available money appropriately; withdrawal/funding functions are capability-gated.

Role	Core wallet use	Withdrawal policy
Tenant / buyer / guest / seeker	Top-ups, available wallet spend, refunded and credited balances, payment/history.	Eligible refundable/available balances only; approved wallet operating model required.
Individual owner / co-owner	Rent/sale/stay allocations, pending proceeds, property-linked history.	Only verified share and payout destination.
Corporate owner / developer / hospitality	Branch/project/asset-level payable and available balances.	KYB, delegated approvers, multi-party approval limits.

Role	Core wallet use	Withdrawal policy
Independent manager / manager company	Contractually earned manager fees; mandate-bound visibility.	Fees only; never owner revenue or owner payout control.
Artisan / service company	Approved deposit, milestone, service payments, pending/available earnings.	Verified business/person, dispute and warranty policies.
Professional / professional firm	Milestone appointment/report fees and reimbursements.	Verification and verified payee for authorized fees.
Admin/platform	Internal operational views; platform revenue and reserve accounts.	Corporate treasury workflow only; not a public user wallet.

2.1 Ledger balance views

View	Definition	Can withdraw?
Payment pending	Customer payment not verified/settled or not yet matched to an obligation.	No
Pending earnings	Provisional or confirmed allocation to a named beneficiary; underlying payment not yet releasable.	No
Protected / held	Funded transaction amount or liability reserved for conditions, security deposit or legal custody.	No
Available	Verified, funded and released amount without applicable freeze or outstanding risk reserve.	Yes, with limits/approvals
Disputed / frozen	Amount subject to dispute, chargeback, fraud, statutory restriction or other permitted hold.	No
Withdrawal pending	Previously available amount now reserved for transfer; actual provider outcome pending.	No
Refund pending	Amount reserved for a refund or remedy; not otherwise spendable.	No

2.2 Accounting controls

- Integer minor units and ISO currency code; currency-specific ledgers; no floating point in financial persistence.
- Immutable balanced journals; compensating transactions for corrections; immutable authorization/policy snapshot for every posting.
- A unified financial identity with separate legal-party, user-role, property/project, funding-provider, beneficiary and transaction accounts.
- Liability-ledger segregation for customer available balances, merchant settlement, protected funds, payout pending, refundable deposits, provider payable and platform revenue.
- Funds coverage = reconciled third-party cash/settled receivable by custody pool minus explicit operational reserves, compared to customer and beneficiary liabilities. No unfunded release.

3. Financial providers and custody architecture

3.1 Provider-neutral roles

Provider capability	Example interface / purpose
Collection	InitializeCharge, VerifyCharge, CancelCharge, ReceiveWebhook
Dedicated virtual account	CreateCustomer, ValidateCustomer, AssignDVA, RequeryDVA, DeactivateDVA
Settlement/balance	FetchSettlement, FetchBalance, ListTransactions, Reconcile
Safeguarding/custody	Reserve/Release/Refund under legally approved custody arrangement; custody proof/reconciliation
Transfer/payout	VerifyBank, CreateRecipient, InitiateTransfer, VerifyTransfer, FinalizeOTP where applicable
Refund/chargeback	InitiateRefund, VerifyRefund, FetchDisputes, FetchReversals
Mandates — later	CreateConsent, AuthorizeMandate, SuspendMandate, AttemptDebit, RevokeMandate

3.2 Preferred Paystack configuration

DESIGN — ENGINEERING: Use Paystack checkout/cards/bank transfer and eligible Paystack-Titan DVAs for collections. Seek Paystack Manual Payouts for corporate merchant balances and Paystack Transfers for beneficiary payouts. Approval and risk acceptance must be obtained first. The preferred flow is Customer → Paystack collection → reconciled Paystack settlement/balance → ledger pending/protected allocation → authorized release → payable/available → Paystack transfer to verified bank.

GATED — EXTERNAL: Manual Payouts govern merchant settlements; they are not equivalent to licensed third-party escrow. Paystack eligibility guidance lists escrow services as unsupported under standard terms. High-value sales, deposits and long-duration custody require documented banking/legal structure, approved provider product and contractual restrictions. Do not launch merely because an endpoint succeeds in test mode.

3.3 Provider routing policy and migration

- Route based on transaction category, geography/currency, verification, legal custody capability, amount, uptime, pricing, liquidity pool, fees and tenant/provider constraints.
- Select and persist route before charging; never auto-retry through another provider while a prior charge or transfer outcome is unknown.
- For a transaction with collected funds at Provider A, payout from Provider B only after explicit approved inter-provider funding, balance reconciliation and migration ledger journal.
- A provider switch affects new transactions by default; in-flight transactions remain bound to original provider/custody pool unless a controlled migration is approved.
- Provider credentials, webhooks, modes and capabilities are per legal entity/environment, securely stored and rotated. Country/currency support is declarative.

3.4 Virtual account provisioning policy

LOCKED — PRODUCT: Every public user gets a wallet record; a Paystack-Titan DVA is provisioned only for eligible users who expressly consent, meet Paystack identity requirements and have a supported business use case. DVA is a collection address, not proof of separately held beneficiary funds. DVA maps one external

identifier to an internal wallet owner and legally appropriate merchant context. Suspended/reassigned accounts require matching and review.

4. Payment lifecycle, beneficiary allocations and disbursements

4.1 Canonical transaction sequence

1. Business service locks a full disclosed quote and agreement/policy version; finance confirms permitted jurisdiction/transaction type.

2. Core creates immutable payment intent and chosen provider route; identifies each charge component and its payer.

3. Paystack/PSP collects; Core confirms signature/event + verified amount/currency/reference/recipient and records clearing versus settled funding.

4. Split engine validates beneficiary authority and computes a versioned allocation snapshot with fee/processing/tax treatment.

5. After financial confirmation, allocations appear pending/protected; no early internal spend or external payout.

6. Owning business service emits validated milestone completion evidence; release engine checks policy, disputes, account ownership and actual funding.

7. Release eligible portions to beneficiary available balances; keep security deposits, future milestones, reserves and disputed amounts separately.

8. Withdrawal locks available funds, uses verified destination, authenticates step-up and risk/approval, then triggers external transfer.

9. Webhook and read-back settle terminal outcome; ledger reservations are released/consumed only on confirmed result; notify all affected parties.

4.2 Allocation invariants

- Gross customer payable = sum(disclosed line items) and must equal accepted quote and collected amount, within explicitly defined permitted adjustments.
- Sum of all beneficiary shares, authorized retained taxes/fees, protected deposits and reconciled processing allocations = payable, with no unexplained residual.
- All beneficiary shares are frozen to the accepted version; changing recipient or percentages requires new approvals and a traceable correction.
- A beneficiary share has independent hold/release/dispute state; one disputed share must not automatically block clean portions if contract/law allows release.
- Co-owner percentages, managers and corporate revenue shares come only from verified ownership/legal mandates and approved beneficiary records.
- No auto-release purely on elapsed time where evidence or legal obligations remain outstanding; timeout actions require a policy and, where needed, human review.

4.3 Wallet example — short-let (illustration only)

Component	Illustrative amount	Ledger/release handling
Host share	NGN 85,000	Pending until relevant stay/booking release condition; separate damage deposit.
Platform service fee	NGN 10,000	Platform revenue only when earned under disclosed policy.

Component	Illustrative amount	Ledger/release handling
Manager fee	NGN 5,000	Pending under verified active mandate and fee terms.
Total	NGN 100,000	All allocations reconcile to confirmed charge; provider processing fees handled per explicit fee policy.

5. Category-specific financial policy catalogue

REQUIRED — SOURCE and DESIGN — ENGINEERING: All listed families get separate policy keys, supported charging schedules, cost components, beneficiary sets, conditions, cancellation/refund and dispute rules. Described triggers below are proposed; exact legally binding timelines and defaults need approved jurisdiction- and contract-specific policy packs.

5.1 Residential initial rent

Payer: Tenant. Components: Rent; one-time legal/onboarding/platform charges; separately refundable deposit; disclosed service/utility charges.

Potential beneficiaries: Owner/co-owners; authorized professional for actual legal work; platform; authorized manager only for contracted fee. Release/cancellation policy: Signed agreement + accepted quote + agreed commencement/possession evidence; deposit retained under separate policy.

5.2 Residential subsequent rent

Payer: Tenant. Components: Current contractual rent and separately lawful agreed recurring items, NOT initial one-time charges.

Potential beneficiaries: Owner/co-owners; platform owner-side commission; authorized manager fee. Release/cancellation policy: Due rent, valid tenancy, compliant rent review/payment terms; no fresh legal fee absent new service.

5.3 Fixed/periodic tenancy renewal

Payer: Tenant. Components: Installment or revised disclosed rent per lawful approved agreement.

Potential beneficiaries: Owner; manager; platform as separately disclosed owner charge. Release/cancellation policy: Continuing tenancy can remain under signed agreement; material changes require appropriate notice/consent/addendum.

5.4 Residential security deposit

Payer: Tenant. Components: Refundable deposit only, separate from rent.

Potential beneficiaries: Tenant refund; owner only for approved evidenced deductions; provider if ordered remedy. Release/cancellation policy: Move-in/out evidence, allowed claims, tenant challenge, dispute resolution and statutory deadlines.

5.5 Commercial rent / lease

Payer: Individual/corporate lessee. Components: Rent; fit-out; service charge; approved refundable deposit.

Potential beneficiaries: Business owner; manager; platform; documented counterparties. Release/cancellation policy: Executed lease, commencement, corporate signatories, escalation and jurisdiction rules.

5.6 Land lease or temporary use

Payer: Land lessee. Components: Reservation, occupation, installments, restoration deposit if agreed.

Potential beneficiaries: Verified landowner(s); platform; appointed professional if fee earned. Release/cancellation policy: Legal use/authority, possession and milestones; separate refundable restoration reserve.

5.7 Residential or commercial purchase

Payer: Buyer. Components: Reservation, deposit, staged/final sale, professional and registrational fees.

Potential beneficiaries: Verified seller(s), professional(s), platform, lawful third parties. Release/cancellation policy: Title/due diligence, closing checklist, conditions precedent, handover and registration logic.

5.8 Land purchase

Payer: Buyer. Components: Reservations, consideration, survey/title/closing expenses.

Potential beneficiaries: Verified landowner(s), surveyor/lawyer if separately instructed, platform. Release/cancellation policy: Verified survey/title, encumbrance review, possession and relevant closing terms.

5.9 Off-plan/developer unit

Payer: Buyer. Components: Reservation, construction installments, handover payment and reserves.

Potential beneficiaries: Verified developer; authorized project parties; platform; professionals. Release/cancellation policy: Independently certified construction milestones, project allocation, snagging/remedies.

5.10 Short-let / serviced apartment

Payer: Guest. Components: Accommodation, cleaning, booking fees, refundable damage deposit, add-ons.

Potential beneficiaries: Owner/host/operator; manager; cleaning provider; platform. Release/cancellation policy: Cancellation/refund plan, check-in/stay completion/checkout; damage reserve separate.

5.11 Hotel / resort

Payer: Guest/corporate booker. Components: Room/board, services, extras, taxes, deposit and penalties if lawful.

Potential beneficiaries: Hotel/operator; service partners; platform. Release/cancellation policy: Rate-plan terms, confirmed folio, check-in/out, authorized extras, no-show policy.

5.12 Event venue / temporary property use

Payer: Event customer. Components: Reservation, event charge, damage/cleaning reserve.

Potential beneficiaries: Owner/operator; service providers; platform. Release/cancellation policy: Event completion, verified cleanup/damages and permitted deductions.

5.13 Maintenance / artisan

Payer: Tenant/owner/business responsible payer. Components: Call-out, materials, labor, milestone fees, approved extras.

Potential beneficiaries: Artisan/company, authorized materials supplier, platform. Release/cancellation policy: Approved quote, access, attendance, evidence, acceptance/rework/warranty policy.

5.14 Professional services

Payer: Buyer/owner/business. Components: Retainer/appointment, milestone/report/visit fees, reimbursable expenses.

Potential beneficiaries: Licensed professional or firm; platform. Release/cancellation policy: Verified mandate/conflict check, signed deliverable, review/acceptance condition.

5.15 Property manager fees

Payer: Owner/business. Components: Per unit, fixed, %, booking, inspection or approved additional service.

Potential beneficiaries: Mandated manager/company; platform if separately earned. Release/cancellation policy: Mandate validity, fee agreement, period/service evidence; never transfer owner revenues to manager.

5.16 Utilities / service charges

Payer: Responsible tenant/owner/business. Components: Verified bills, shared-meter allocation, approved service budgets.

Potential beneficiaries: Actual utility/service supplier, authorized operator, platform if separately agreed. Release/cancellation policy: Invoice/bill verification, service obligations, challenge and adjustment.

5.17 Promoted listings / subscriptions

Payer: Owner/business. Components: Advert or subscription disclosed fee.

Potential beneficiaries: Platform and contracted marketing provider if any. Release/cancellation policy: Publish/service period, cancellation, refund and accounting recognition.

5.18 Mortgage / insurance / financed payments

Payer: Customer/financing or insurance partner. Components: Premiums/loan installments and fees only when licensed partner approves.

Potential beneficiaries: Licensed partners and expressly eligible counterparties. Release/cancellation policy: Partner contracts and approvals; not Core self-originated lending/insurance.

6. Residential rent — locked pricing and contract logic

6.1 Initial vs recurring charge rule

LOCKED — PRODUCT: One-time agreement preparation, onboarding and first-time platform charges are not repeated solely because the next rent installment is due. Normal recurring rent is not automatically accompanied by a new legal or agreement fee. Genuine new legal service may be charged only when actually instructed, legally permissible, quoted and accepted.

Illustrative original invoice	Amount	Illustrative next annual invoice	Amount
Annual rent	NGN 1,000,000	Contractual rent	NGN 1,000,000
Combined one-time platform/legal charges	NGN 200,000	Recharged initial legal/platform fees	NGN 0
Initial total (excluding any distinct deposit)	NGN 1,200,000	Recurring tenant payment (absent other agreed items)	NGN 1,000,000

OPEN — BUSINESS: The composition of the initial NGN 200,000 is an illustration from stakeholder discussion, not an approved tariff. Any security deposit, tax or service charge must be separately disclosed, not assumed included.

6.2 Platform recurring revenue

LOCKED — PRODUCT: The preferred standard is a transparently contracted owner-side platform service commission deducted from the owner's rent proceeds or a documented owner subscription. The tenant should not face repetition of original onboarding/legal fees. The legal fee is payable only for actual agreed legal work.

Illustration only: on NGN 1,000,000 annual rent, a hypothetical 2% owner-side fee would allocate NGN 20,000 to CosmicForge and NGN 980,000 to the owner before other contracted manager fees and provider charges. The 2% is NOT an approved or hardcoded rate.

6.3 Rent cycle vs contract continuity

- Fixed-term tenancy and continuing/periodic tenancy must both be supported. A rent installment is not the same as execution of a fresh agreement or renewal of a fixed term.
- A continuing agreement may govern recurring rent until changed or terminated under applicable law; retain versioned original terms, permitted rent reviews, notice periods and obligations.
- Where law/contract requires fresh agreement or addendum for a material change, use reviewed templates and approvals. No automatic legal charge without service and consent.
- Payment schedule may be annual, semiannual, quarterly, monthly or another lawful agreed interval; the base rent obligation has an immutable reference and due date.
- Recurring wallet auto-pay requires opt-in and revocable mandate, but execution of scheduled payment instructions and saved-bank-account charging remain DEFERRED.

6.4 Sample rent ledger event sequence

LeaseActive → RentObligationDue → WalletFundingOrCheckout → PaymentVerified → RentSplitPending → ReleaseEvidenceAccepted → OwnerCommissionAllocated → OwnerAvailable + PlatformEarned → ReceiptsAndStatement. Deposit logic is a separate linked but non-nettable ledger stream.

7. Refundable deposits, reserves, disputes and cancellations

7.1 Deposit segregation

Security deposits, hospitality damage deposits and restoration deposits have distinct ledger liabilities, payout rules and custody references. They cannot be treated as rent, platform revenue or owner available earnings merely when received. Record ingress, inspection/condition evidence, deduction claim and tenant challenge separately.

7.2 Dispute workflow

- Dispute intake binds payment, beneficiary shares, property/job/booking, authority, documentary evidence, deadline and review officer.
- Freeze only contested amounts where feasible; allow uncontested portions to progress under approved policy.
- Decisions require separated case-worker and payout authority where thresholds dictate; every override logs reason, signer and evidence.
- Support outcome: release to entitled party, return to payer, partial award, reserve extension, rework or legal escalation.
- Cases and property records remain linked after account closure for lawful retention; explicit statutory deposit refund deadlines and escalations must be configured by jurisdiction.

7.3 Refund, chargeback and reversal correctness

- Track every refund attempt independently, allow multiple partial refunds with cumulative caps and race protection; do not prematurely mark original payment refunded.
- Reversed/charged-back charges must correctly adjust external funding, liabilities, recoverable amounts and reserves without deleting financial history.
- Refund provider results of timeout/unknown outcome remain reserved pending read-back; never issue blind duplicate refund or a balance refund at the same time.
- Refund destination must obey instrument and legal/provider rules; wallet refund is not always interchangeable with original card/bank reversal.

8. Provider payouts and safe available balances

8.1 Verified beneficiary and mandate

- Individual owners: beneficiary must match legal authority or documented verified payout authority; co-owner shares follow signed entitlement and consent.
- Manager payouts derive only from an active, property-specific management mandate and fee schedule; managers cannot redirect owner disbursements.
- Business payouts follow corporate identity, branch/project mapping, approved bank destination, financial approval chain and limits.
- Artisan and professional payouts go to the verified individual/firm named on completed work or signed appointment, not to arbitrary staff.
- Change of payout bank requires step-up, identity revalidation, cooling-off and notification; high-value actions can require dual approval.

8.2 Payout state machine

REQUESTED → RISK_REVIEW / MANUAL_APPROVAL (if configured) → RESERVED → SUBMITTED → PROCESSING / OTP_REQUIRED / UNKNOWN_OUTCOME → CONFIRMED_PAID or CONFIRMED_FAILED / REVERSED. No terminal outcome without verified external evidence; never restore funds on a mere timeout. All retries preserve provider transfer reference and idempotency.

8.3 Liquidity safety rule

AvailableBalanceDisplayed can be positive only when the policy-authorized amount is recorded as earned and backed by reconciled custodial or settlement funds (or an explicitly authorized prefunding arrangement). Actual withdrawal may wait for settlement/rail availability; show separate status instead of false immediate payout.

9. Customer funding and future recurring rent automation

9.1 Core customer top-up and spend

- Customer can initiate an approved wallet top-up via PSP checkout or assigned DVA; wallet credit occurs only after verified amount/reference/customer attribution.
- Wallet payment of accepted rent/booking/service obligation is atomic and idempotent; debit customer available and credit transaction protected liability.
- Available customer funds are not automatically treated as earning for owners; settlement and business release conditions still apply.
- Customer wallet withdrawal/refund capabilities are separate and must not be shown as operational until provider/compliance approval and backend certification.

9.2 Recurring payment design reservation — DEFERRED

Future work will support an opt-in autopay mandate on available wallet balance, low-balance notifications, installment scheduling, payment retries and escalation. A stored bank account must NEVER be charged without an authorized mandate and compliant debit rail. Saved-card authorization and direct-debit provider integrations are explicitly out of scope for core v1 and must be separately reviewed. Do not implement fake or client-side simulated success.

10. Financial states, posting invariants and reliability

10.1 Payment intent lifecycle

CREATED → REQUIRES_ACTION / PENDING_PROVIDER → VERIFIED_COLLECTED → AWAITING_SETTLEMENT → FUNDED_PROTECTED / FAILED / CANCELLED / UNKNOWN_OUTCOME. Partial capture/partial funding requires explicit schema and policies. Provider webhooks alone are inputs, not indisputable authorization.

10.2 Beneficiary allocation lifecycle

PROPOSED → LOCKED → PENDING_FUNDING → FUNDED_HELD → PARTIALLY_RELEASED / AVAILABLE → WITHDRAWAL_RESERVED → PAID; alternative DISPUTED / REFUND_RESERVED / CANCELLED states. Allocation cannot become AVAILABLE without policy + sufficient backing + verified beneficiary.

10.3 Durability and operational safeguards

- Transactional outbox/inbox, signature validation, event deduplication without suppressing retry of failed events, bounded backoff and dead-letter queues.
- Unknown external outcomes recover by read-back, not by cancellation; idempotency keys persist original request fingerprint and final outcome.
- Row locks/advisory locks for balances/transactions; at-most-once economic effects despite at-least-once delivery; no phantom success.
- Double-entry integrity checked per currency/legal entity; payout/refund transfer references unique and reconciled.
- Backups, point-in-time recovery, secrets rotation, service-to-service authorization, audit retention, alerting and incident response.
- Daily automatic reconciliation plus intraday checks when needed; provider balance, pending settlements, fees, chargebacks, reserves, liabilities and unmatched credits.

11. Domain model — implementation-level entity catalogue

Entity / aggregate	Core fields / invariants
FinancialIdentity	party_id, legal_entity, user_roles, KYB/KYC state, activation; immutable cross-service identity mapping.
Wallet / LedgerAccount	wallet_id, owner, account_type, currency, legal_entity, custody_pool, balances derived from journals.
Journal / JournalEntry	journal_id, idempotency fingerprint, account, side, amount_minor, currency, source, status, audit.
Provider / ProviderCapability	provider_id, approved merchant, currency/market, collection/custody/payout capabilities, activation state.
ProviderAccount / VirtualAccount	customer mapping, external ref, consent, assign/status, bank code, masked account, validation expiry.
PaymentIntent / ChargeAttempt	actor, obligation, full-price snapshot, amount, method, provider route, external ref, verification.
Settlement / CustodyPool	provider, legal account, settlement batch, balance, source of funds, liquidity availability, restrictions.
Quote / ChargeLine	rent, fees, taxes, deposit, service charges, refundability, one-time/recurring flag, source authority.
AllocationPlan / BeneficiaryShare	versioned share basis, beneficiary, entitlement evidence, percentage/fixed, net/gross, release profile.

Entity / aggregate	Core fields / invariants
ReleasePolicy / ReleaseRequest	jurisdiction, type, conditions, evidence, approvals, frozen amounts, earliest lawful release.
Agreement / ObligationRef	linked business agreement, contract version, periodic rent, due dates, unchanged fee fingerprint.
DepositLiability / DeductionCase	deposit purpose, refundable principal, evidence, claims, tenant responses, deadline and decision.
Dispute / FinancialHold	case, contested shares, authority, status, evidence, expiry and authorized disposition.
PayoutAccount / Transfer	verified payee, masked bank, cooling-off, approval, reference, state, retries and finality.
RefundAttempt / Chargeback	original collection ref, cumulative refundable amount, reserved amount, provider outcomes, ledger offsets.
Invoice / Receipt / Statement	party, transaction, fee-line evidence, versions, tax, signed receipt, export.
WebhookInbox / Outbox / Reconciliation	signature, event status, failure/retry, external provider statements, unmatched exception.
MandatePlaceholder (deferred)	customer opt-in, scopes, scheduling and revocation contract only; no live direct debit in v1.

11.1 Minimum invariant set

1. Every posted journal balances separately by currency and legal entity.
2. No ledger debit can make a restricted available balance negative absent an explicitly approved credit facility.
3. No amount may simultaneously be both protected and freely available to the same beneficiary.
4. Per-transaction beneficiary allocations reconcile exactly to verified line items and approved adjustments.
5. Cumulative captures, refunds, releases and payouts never exceed their respective funded or entitled amounts.
6. Provider funds never silently bridge legal entities, currencies, custody pools or in-flight transfers.
7. An approved bank-change cooldown or dispute blocks relevant withdrawals.
8. Cash/settlement and customer liability coverage exceptions immediately produce holds and operator alerts.

12. Contracts, APIs, events and service integration map

DESIGN — ENGINEERING: Names below are canonical contract proposals and must be reconciled with the real repository before adoption. OpenAPI schemas, authorization and event versions take precedence over code snippets or endpoint-name guesses.

12.1 Payment Service API families

Surface	Example operations (proposal)	Caller / condition
Customer wallet	GET /wallets/me; GET /transactions; POST /topups; POST /topups/{id}/verify	Authenticated payer; verify ownership and idempotency.
Funding/quote	POST /payment-intents; GET /payment-intents/{id}; POST /wallet-payments	Business service or restricted payment orchestration.
Beneficiary earnings	GET /wallets/me/earnings; GET /allocations/{id}	Authorized owner, manager, artisan, professional or company context.

Surface	Example operations (proposal)	Caller / condition
Beneficiary payouts	GET/POST /payout-accounts; POST /withdrawals; GET /withdrawals/{id}	Verified payee; step-up, mandate, cooling-off and risk policy.
Release controls	POST /internal/releases/request; GET /internal/releases/{id}	Source-bound HMAC/mTLS; milestone evidence, policy checks.
Dispute/deposit	POST /internal/holds; POST /internal/deposit-decisions; POST /internal/refunds	Authorized support/finance/business service; dual-control rules.
Provider integration	POST /webhooks/{provider}; GET /internal/provider-status/{ref}	Signature/auth verified and durable processing.
Finance operations	GET /admin/reconciliation; POST /admin/freeze; POST /admin/approve-withdrawal	Least privilege; maker-checker where required.

12.2 Canonical event catalogue (proposal)

- finance.wallet.opened.v1; finance.virtual_account.assigned.v1; finance.wallet.topup_verified.v1
- finance.payment.intent_created.v1; finance.payment.collected.v1; finance.payment.funded.v1; finance.payment.unknown_outcome.v1
- finance.allocation.locked.v1; finance.allocation.pending.v1; finance.allocation.released.v1
- finance.deposit.held.v1; finance.deposit.deduction_proposed.v1; finance.deposit.refunded.v1
- finance.dispute.hold_applied.v1; finance.dispute.resolved.v1
- finance.refund.requested.v1; finance.refund.completed.v1; finance.chargeback.recorded.v1
- finance.withdrawal.requested.v1; finance.withdrawal.under_review.v1; finance.withdrawal.completed.v1; finance.withdrawal.failed.v1
- finance.reconciliation.exception_detected.v1; finance.reconciliation.closed.v1

12.3 Business service integration strategy

Real Estate Individual Owner is the first public user-domain backend. Future Marketplace, Business, Manager, Artisan, Professional and Admin services are not assumed to exist today. The Payment Core must publish durable documented contracts, locally test stubs/adapters and deferred integration registers. Missing future services are NOT defects of the core; unsupported fake success responses are defects.

13. Provider approval, licensing and consumer protection gates

13.1 Paystack-specific gating items

Gate	Needed proof before live use
Business activation	Nigerian registered merchant approved and assigned appropriate business classification.
Manual Payouts	Written enrollment and activation of manual merchant settlement, approved allowed balance use and settlement arrangements.
Paystack-Titan DVA	Merchant feature access, per-user consent/KYC, limits, account owner designation, allowed funding use and reconciliation.
Wallet/top-up	Paystack and licensed partner permit our customer stored-

Gate	Needed proof before live use
	value/use and withdrawals for all targeted user categories.
Protected funds/escrow	Written approval or alternative regulated trustee/custodian model for third-party conditional-release business.
High-value real estate	Provider approval/limits for land, sales, off-plan and large deposits; unusual activity/source-of-funds controls.
Payouts	Transfers enabled, beneficiary verification, limits, approval/OTP requirements and status read-back.
International/diaspora	Distinct approved provider/risk framework; do not assume Paystack international eligibility for real estate.
Legal jurisdiction	Nigeria state-specific rent, deposit, escrow, legal fee, taxes, title-transfer and termination policy packs approved by counsel.

13.2 Conditions that must not be implied

- Paystack merchant balance is not a trust account simply because it can be manually disbursed.
- An in-app "wallet" does not by itself authorize CosmicForge to issue redeemable stored-value, hold customer deposits or operate money transmission.
- A tenant prepaid annual rent is not automatically lawful to hold for an arbitrary period; follow actual contract and local law.
- The ledger can reserve pending amounts even before custody is live, but must not label them "safeguarded" or "escrow protected" unless the corresponding custody arrangement exists.
- No shared Paystack balance funding model may use one customer/transaction's restricted funds to pay a different party.

14. Engineering delivery workstreams

Workstream	Scope	Evidence expected
F1 — Discovery/authority	Map existing repo, shared foundations, payment contracts, source trace register, legal gates.	Inventory and no-duplication plan.
F2 — Ledger/wallets	Financial identity, wallet accounts, immutable journals, consistent balances and account permissions.	DB/race/migration and invariant tests.
F3 — Provider adapters	Provider routing framework, Paystack checkout, verification, webhooks, manual-balance reconciliation and DVA.	Provider contract tests and sandbox traces.
F4 — Quotes and splits	All-in cost lines, one-time vs recurring flags, split engine, beneficial ownership and fee rules.	Cross-category allocation fixtures.
F5 — Custody and release	Protected pools, per-transaction milestone evidence, partial release and disputed holds.	Policy replay/evidence authorization tests.
F6 — Transfers/refunds	Payout accounts, approvals, OTP where supported, unknown outcome, refunds and chargebacks.	Negative/adversarial case suites.

Workstream	Scope	Evidence expected
F7 — Recurring rent obligations	Installment ledger, rent due data contract, customer wallet payment (manual); deferral hooks for auto-pay.	Initial vs recurring rent regression.
F8 — Finance/admin	Statements, approvals, risk, fraud cases, reconciliation dashboards and audit.	Daily balance coverage certification.
F9 — Cross-service certification	Owner live/stub integration first; future services contract-driven.	Real DB, Paystack test environment, E2E and security gates.
F10 — Staging/go-live	Deploy, alerting, provider approval, legal policy activation and production sign-off.	Signed readiness/certification packet.

14.1 Hard gates for every workstream

- Stop or mark blocked on missing authoritative provider documentation or business policy; never invent provider abilities, legal authority or external service implementations.
- Test with real PostgreSQL, test provider accounts and event callbacks; execute concurrency and unknown-outcome scenarios, not only mocked unit tests.
- No production secrets in the repository; no development fallback secrets in staging/production; RBAC by source service and legal entity.
- Record exact file paths, migrations, OpenAPI/Postman, event schemas, test results, commits and unresolved dependencies per workstream.
- Do not remove, duplicate or overwrite owner-side payment/domain logic already implemented. Integrate it using explicit service contracts.

15. Acceptance scenarios and required test evidence

Case	Expected invariant / externally visible behavior
A01: Tenant initial rent	NGN 1,200,000 illustrated as NGN 1m rent + NGN 200k distinct one-time lines; correct authorized beneficiaries and separate deposit if any.
A02: Tenant subsequent rent	Only NGN 1m rent due absent separate lawful recurring items; one-time fee not repeated; any owner commission deducted only per contract.
A03: Continuing tenancy	Next rent payment does not force fresh agreement or legal invoice; revision/termination require lawful agreement process.
A04: Multiple roles	Same identity can be customer and owner; ledger/source permissions remain separated.
A05: Multi-beneficiary short-let	Correct owner/manager/platform allocations; each pending until relevant milestone; release separately.
A06: Split dispute	Freeze disputed share and permit compliant release of undisputed shares.
A07: Co-owned property	Verified co-owner percentages/mandate control beneficiary shares; unapproved manager cannot redirect.
A08: Deposit claim	Deposit protected separately; itemized evidence/deduction/challenge/partial refund and deadlines work.
A09: Off-plan milestones	Cannot release uncertified construction installment; reversal/withdrawal subject to project contract.

Case	Expected invariant / externally visible behavior
A10: Payment timeout	Unknown external charge status; read-back before alternate provider route. Never double-charge.
A11: Webhook failure	Failed event retried durably; duplicate webhook creates no duplicate credit.
A12: Transfer timeout	Withdrawal balance remains reserved until confirmed; no double payout.
A13: Failed transfer	State guards compensate precisely once; final balance restored only on definitive failure.
A14: Payout account change	Strong verification, cooldown, notification and high-value approvals enforced.
A15: Provider swap	New transaction routes change; in-flight funds remain correctly assigned to original provider/custody pool.
A16: Accounting proof	Balanced entries + provider settlement/custody reconciliation show adequate coverage for all liabilities.
A17: No Paystack approval	Protected buyer-seller escrow and unrestricted wallets remain non-production; safe unavailability clearly displayed.
A18: Role withdrawal	All eligible receiving roles can withdraw only their verified available share; customer refunds/withdrawal subject to approval.
A19: DVA consent	No account issued by default without recorded consent and provider validation.
A20: Deferred autopay	No saved bank charge occurs; future mandate fields do not trigger live debit.
A21: Source isolation	One business service cannot complete another service's transaction or release its funds.
A22: Reconciliation exception	Money mismatch freezes risky releases and creates audited finance-ops queue with resolution.

16. Decision register, pending approvals and change control

16.1 Locked decisions

Decision ID	Approved product baseline
D-001	Every user-role account has an internal wallet identity; eligible money visibility and withdrawal are role/capability governed.
D-002	One Payment Core owns ledger and all split/release/refund/withdrawal accounting.
D-003	Multiple providers through adapters; Paystack first where commercially and legally approved.
D-004	Manual Paystack settlements preferred; no routine bounce from company operating bank to Paystack.
D-005	Held allocations visible to users but unavailable for withdrawal until respective release conditions.
D-006	All-in disclosure and beneficiary-specific split and release rules per transaction family.

Decision ID	Approved product baseline
D-007	No recurring initial legal/onboarding/platform fees on normal rent payments.
D-008	Preferred owner-paid recurring platform commission or agreed owner subscription; no undisclosed deductions.
D-009	Support fixed-term and continuing rental agreements; payment of the next installment does not necessarily require new contract.
D-010	Core top-ups and manual wallet payments now; saved-bank/card automatic charging later.
D-011	Future user-domain services contract-first; do not treat their absence as incomplete current implementation.
D-012	Regulatory and provider-approved custody is a hard live-money gate.

16.2 Pending business/legal decisions

Open ID	Decision needed / owner
O-001	Exact platform commission by product, payer and jurisdiction; pricing committee/finance.
O-002	Who bears PSP processing fees, withholding/tax and failed-transfer charges; finance/legal.
O-003	Exact rent release milestone: possession, signed agreement, grace window, verified handover; legal/product.
O-004	Per-state fixed-term/periodic agreement templates, deposit/notice/advance-rent restrictions; Nigeria counsel.
O-005	Per-category refund/no-show/cancellation/release deadlines and dispute escalation; product/legal.
O-006	Lawful custody/escrow structure; banking/legal/compliance + Paystack written approval.
O-007	Payout thresholds, dual approval, reserve ratios, co-owner rules and provider limits; finance/risk.
O-008	DVA provisioning policy, customer withdrawal permission, Paystack capacity and consents; payments/compliance.
O-009	International/diaspora PSP and permitted cross-border payments; compliance/providers.
O-010	Service boundary: Real Estate-only initial deployment vs future CosmicForge-wide financial platform; architecture.
O-011	Accepted fee/legal invoicing workflows for genuine new agreement vs simple recurring rent; legal/product.
O-012	Exact settlement SLA and liquidity reservation policy by provider; treasury/providers.

16.3 Change control and source precedence

- Every change gets decision ID, rationale, impacted source chapter, migration/contract impact, tests, migration strategy, approval authority and effective date.
- No change to legal custody, recurring platform fees, beneficiary entitlement or deposit rules without explicit approval and versioned customer disclosure.
- This blueprint is a shared platform architecture standard; domain owners remain authoritative for legal business facts and evidence.
- Mark implementation statuses individually: NOT STARTED, IMPLEMENTED LOCALLY, CONTRACT PASS, E2E PASS, STAGING PASS, PRODUCTION APPROVED, EXTERNAL BLOCKED.

17. Source-to-requirement traceability

Requirement	Product documents	Sections in this blueprint
Full cost transparency, no hidden charges	00 §10; 01 §20; 02 §8; 07 §10	§4, §5, §6
Protected money, staged release and split payment	00 §11; 01 §16/§20; 03 §4/§12; 07 §11	§3–§5, §7, §10
Security deposits, deductions, disputes	00 §12/§23; 01 §11/§20; 02 §14–15; 07 §12/§15	§5, §7
Owner/beneficiary verification	00 §6/§8; 02 §3–4; 03 §4–6	§2, §4, §8
Corporate multi-party financial controls	03 §3–4; 07 §1/§11	§2, §8, §13
Manager mandate and fee limitations	04 §5/§17; 02 §21; 03 §19	§2, §5, §8
Artisan milestones and invoices	05 §6/§8/§13	§5, §7
Professional fees and reports	06 §4/§12/§16	§5, §7
Reporting and reconciliation	00 §11; 03 §4; 07 §11/§25	§8, §10, §15
Rent installments and lawful renewal	01 §11/§20; 02 §14–15; 03 §9	§6, §9
Multi-provider, Paystack Manual Payouts, every-user wallets	Stakeholder approved design additions — not explicitly specified in product v5.0	§2, §3, §16
Owner-side recurring commission instead of repeated legal fees	Stakeholder approved design addition — not fixed in product v5.0	§6, §16

18. External-provider reference register

Current provider documentation consulted on 10 October 2026. This register is contextual product research, not an approved commercial agreement.

Paystack Manual Payouts — https://support.paystack.com/en/articles/2131074

Registered merchant settlement into merchant Paystack balance; requires approval.

Paystack Dedicated Virtual Accounts — https://paystack.com/docs/payments/dedicated-virtual-accounts/

Explicit user consent and applicable identity requirements; collections to merchant.

Paystack Transfers — https://paystack.com/docs/transfers/single-transfers/

Recipient creation, transfer reference, initiation and terminal status.

Paystack Transfer Recipients — https://paystack.com/docs/transfers/creating-transfer-recipients/

Verified external recipient creation.

Paystack Split Payments — https://paystack.com/docs/payments/split-payments/

Automatic settlement splitting; NOT our protected-release mechanism.

Paystack Webhooks — https://paystack.com/docs/payments/webhooks/

HMAC signature, asynchronous events and durable processing.

Paystack Verify Payments — https://paystack.com/docs/payments/verify-payments/

Backend read-back and exactly-once digital value fulfillment.

Paystack Ineligible Businesses — https://support.paystack.com/en/articles/2127042

Standard terms list escrow services as unsupported.

Paystack International Eligibility — https://support.paystack.com/en/articles/2127682

Real estate listed under international-payment ineligible categories.

Final architecture sign-off statement

This document locks the user-approved product architecture for a centralized, provider-neutral CosmicForge Real Estate Payment Service. It explicitly includes every public user wallet, internal beneficiary allocations, transparent multi-party splits, milestone-based release, deposit separation, verified withdrawals, tenant top-ups, distinct initial and recurring rent costs, continuing tenancy support and deferred saved-bank auto-debit.

No software implementation or live financial activation is authorized merely by this document. Before coding, the engineering agent must audit the target repository and write a source-to-contract coverage plan. Before live money, the business must obtain provider acceptance, lawful custody/stored-value permissions, state-specific policy approval, verified settlement arrangements and real financial E2E certification.

END OF CONTROLLED MASTER BLUEPRINT — COSMICFORGE REAL ESTATE FINANCIAL CORE v1.0